

Assignment 1 : due 14 Aug before class

1. Calculate the size of the quanta involved in the excitation of (a) an electronic motion of period 1.0 fs, (b) a molecular vibration of period 10 fs, and (c) a pendulum of period 1.0 s.

Ans. Use $E = h\nu$; $h = 6.626 \times 10^{-34}$ J s; $\nu = 1/T$ [T : period]

(a) $E = 6.626 \times 10^{-34} \text{ J s} / 1.0 \times 10^{-15} \text{ s} = 6.626 \times 10^{-19} \text{ J}$

(b) $E = h / (1.0 \times 10^{-14} \text{ s}) = 6.626 \times 10^{-20} \text{ J}$

(c) $E = h / (1.0 \text{ s}) = 6.626 \times 10^{-34} \text{ J}$

2. Deduce the Eqn. $\lambda - \lambda_0 = \frac{h}{m_e c} (1 - \cos \theta)$ for the Compton effect on the basis of the conservation of energy and linear momentum. Use the relativistic expressions. Initially the electron is at rest with energy $m_e c^2$. When it is travelling with momentum p its energy is $\sqrt{p^2 c^2 + (m_e c^2)^2}$. The photon, with initial momentum $\frac{h}{\lambda_0}$ and energy $h\nu$, strikes the stationary electron, is deflected through an angle θ , and emerges with momentum $\frac{h}{\lambda}$ and energy $h\nu'$. The electron is initially stationary but moves off with an angle ϕ to the incident photon. Conserve energy and both components of linear momentum.

The quantity, $\frac{h}{m_e c}$ is called the Compton wavelength of the electron (2.426 pm).



Ans. (before) (after)

For convenience, let us write quantities for electron with the label 'e' and for the photon with no label. Subscripts 'i' and 'f' are for initial and final.

Conservation of energy $\Rightarrow E_i + E_{e_i} = E_f + E_{e_f}$

or, $h\nu_i + m_e c^2 = h\nu_f + \sqrt{p_e^2 c^2 + (m_e c^2)^2}$

or, $p_e^2 c^2 = (h\nu_i)^2 + (h\nu_f)^2 - 2h^2 \nu_i \nu_f + 2m_e c^2 (h\nu_i - h\nu_f)$ ——— (A)

Conservation of momentum $\Rightarrow \vec{p}_{p_i} = \vec{p}_{p_f} + \vec{p}_e$

or, $\vec{p}_e = \vec{p}_i - \vec{p}_f$

and $p_e^2 = (\vec{p}_i - \vec{p}_f) \cdot (\vec{p}_i - \vec{p}_f) = p_i^2 + p_f^2 - 2p_i p_f \cos \theta$

or, $(cp_e)^2 = (h\nu_i)^2 + (h\nu_f)^2 - 2h^2 \nu_i \nu_f \cos \theta$ ——— (B)

From Eqns. A and B, obtain,

$-2h^2 \nu_i \nu_f \cos \theta = -2h^2 \nu_i \nu_f + 2m_e c^2 (h\nu_i - h\nu_f)$

and this gives $\frac{1}{h\nu_f} - \frac{1}{h\nu_i} = \frac{1}{m_e c^2} (1 - \cos \theta) \Rightarrow \lambda - \lambda_0 = \frac{h}{m_e c} (1 - \cos \theta)$

3. Which of the following functions are eigenfunctions of (a) $\frac{d}{dx}$, (b) $\frac{d^2}{dx^2}$:

(i) e^{ax} , (ii) e^{ax^2} , (iii) x^2 , (iv) $\sin x$?

Ans. (a) (i), and (b) (i) and (iv) (explanation needed)

4. A crude treatment of the pi electrons of a conjugated molecule regards these electrons as moving in the particle-in-a-box potential, where the box length is somewhat more than the length of the conjugated chain. For butadiene, $\text{CH}_2=\text{CHCH}=\text{CH}_2$, take the box length as $7.0 \text{ \AA}$ and use this model to estimate

the wavelength of light absorbed when a π -electron is excited from the highest-occupied to the lowest-vacant level. The experimental value is 217 nm.

Ans. Butadiene has 4 π -electrons. They will fill 2 MOs. The excitation on absorption of light involves energy, $\Delta E_{3 \leftarrow 2} = \frac{(3^2 - 2^2)h^2}{8m_e L^2}$.

$$\lambda = \frac{hc}{\Delta E} = \frac{hc}{\frac{5h^2}{8m_e L^2}} = \frac{8cm_e L^2}{5h}$$

We can use atomic units, $h = 2\pi, m_e = 1, L = \frac{7}{0.529}$ and $c = 137$

Then, $\lambda = \frac{8 \times 137 \times 49}{5 \times 2 \times \frac{7^2}{0.529^2}} \text{ a.u. of length} = \frac{8 \times 137 \times 49}{5 \times 2 \times \frac{7^2}{0.529^2}} \times \frac{1}{10} \text{ nm} = 322.1 \text{ nm}$

5. Consider a particle in a 1-D box that extends from 0 to L . Assume the particle is in the lowest level of the box. At $t \rightarrow 0$ the size of the box is suddenly expanded, so that the box extends from 0 to $2L$. Assume the particle is in the energy lowest level.

(i) Write the wavefunction at $t = 0$ for the initial box, $\Psi(x, 0)$. Schematically draw a profile of $\Psi(x, 0)$ for the expanded box.

(ii) Write the wavefunctions, $\psi_n(x)$, and the eigenvalues, E_n , for the eigenstates with quantum number n for the expanded box.

(iii) Show graphically that for even $n > 2$, $\Psi(x, 0)$ is orthogonal to all $\psi_n(x)$, but not for $n = 2$.

(iv) Expand $\Psi(x, 0)$ in a complete set of the new eigenfunctions, $\psi_n(x)$, and obtain the amplitudes, a_n .

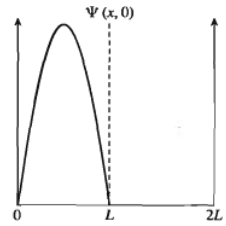
(v) Show by direct sum that $\sum_{n=1}^9 |a_n|^2 \approx 1$.

(vi) Write the expression for $\Psi(x, t)$.

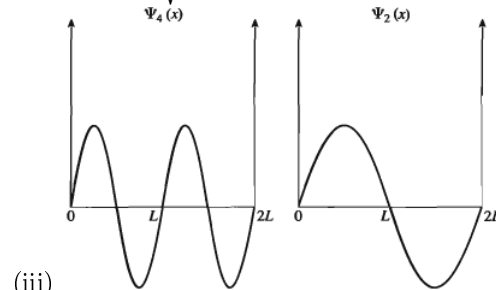
(vii) Find the value of τ if $\Psi(x, \tau) = \Psi(x, 0)$?

(viii) Write the expression for $\Psi(x, \frac{\tau}{2})$.

Ans. (i) $\Psi(x, 0) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$



(ii) $\psi_n(x) = \sqrt{\frac{1}{L}} \sin\left(\frac{n\pi x}{2L}\right)$; $E_n = \frac{n^2 \pi^2 \hbar^2}{8mL^2}$; $n = 1, 2, \dots$



(iii)

$\Psi(x, 0)$ is orthogonal to all $\psi_n(x)$ if n is even. This is because $\Psi(x, 0)$ vanishes outside of the range $[0, L]$ and in that range all the even eigenstates of the full box correspond, within a normalization factor, to excited eigenstates of the half box. This is shown in the left panel for $n = 4$. The one exception is the $n = 2$ state, shown in the right panel, which is essentially a replica of $\Psi(x, 0)$ in the range $[0, L]$.

$$(iv) \Psi(x, 0) = \sum_n a_n \psi_n(x),$$

$$\text{where } a_n = \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \cdot \sqrt{\frac{1}{L}} \sin\left(\frac{n\pi x}{2L}\right) dx$$

From (iii) above, we already know that $a_{n>2} = 0$ if n is even.

$$a_2 = \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) \cdot \sqrt{\frac{1}{L}} \sin\left(\frac{\pi x}{L}\right) dx = \frac{1}{\sqrt{2}}$$

$$\text{For } n \text{ odd, use the formula } \int_0^L \sin(bx) \cdot \sin(cx) dx = -\frac{1}{2} \left[\frac{\sin[(b+c)a]}{b+c} - \frac{\sin[(b-c)a]}{b-c} \right]$$

$$\text{This gives, } a_n = \frac{4\sqrt{2}(-1)^{\frac{n+1}{2}}}{(n+2)(n-2)\pi}$$

$$(v) \sum_1^9 |a|_n^2 \approx 0.6 + 0.707 + 0.36 + 0 - 0.086 + 0 + 0.04 + 0 - 0.023 \approx 1$$

$$(vi) \Psi(x, t) = \sum a_n \sqrt{\frac{1}{L}} \sin\left(\frac{n\pi x}{2L}\right) e^{-i \frac{n^2 \pi^2 \hbar^2}{8mL^2} t}$$

$$(vii) \text{ Say, } \omega_1 = \frac{\pi^2 \hbar^2}{8mL^2}. \text{ Then, when } \omega_1 t = 2\pi, \Psi(x, t) = \Psi(x, 0) \implies \tau = \frac{2\pi}{\omega_1}.$$

$$(viii) \Psi(x, \frac{\tau}{2}) = \sum a_n \sqrt{\frac{1}{L}} \sin\left(\frac{n\pi x}{2L}\right) e^{-in^2 \omega_1 \cdot \frac{\pi}{\omega_1}} = \sum a_n \sqrt{\frac{1}{L}} \sin\left(\frac{n\pi x}{2L}\right) e^{-in^2 \pi}$$